Mobility in 5G Networks

5G-4G Handover – Refer following link

5G To 4G Interworking: Handover Call Flow - NetworkBuildz

Mobility Tracking

- To establish faster connectivity, UE mobility needs to be tracked
- UE informs whenever it moves out of an “Area”
- “Area” needs to be defined by Network
- When network needs to reach out to UE, Pages in the “Area”
- “Area” should be big enough to avoid frequent mobility updates from UE, but small enough to avoid Paging bandwidth wastage
- Can be per UE defined, based on UE movement behavior
- Also optimized for group of UE’s to prevent collective mobility registration

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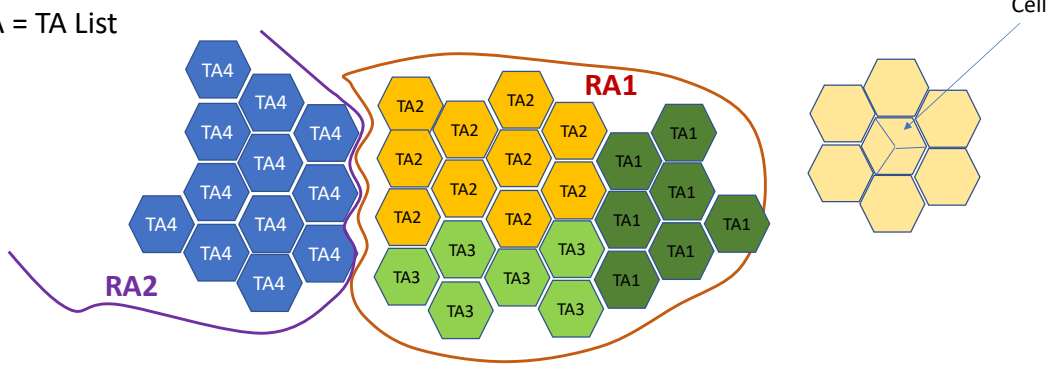
Areas for tracking a UE

- Cell, Tracking Area (non-overlapping), Registration Area
- TAI = Mobile Country Code (MCC) + Mobile Network Code (MNC) + 3-byte TAC
- RA = TA List

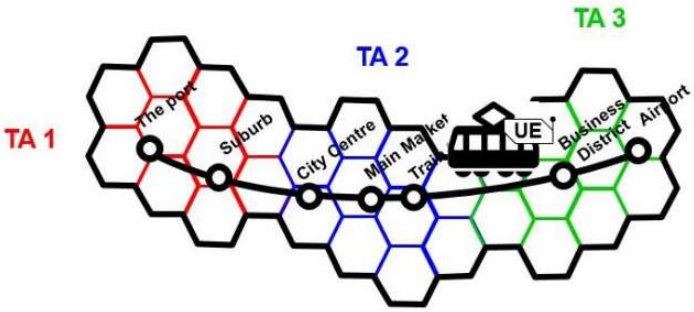


- RAN notification area (in RRC Inactive) – Cells, TAs, RAN areas
- Service Area – Allowed Tracking areas

Dynamic Registration Area

Registration Area

TA List = TA1, TA2, TA3



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Registration Area

- Introduced in 5G equivalent to Location Area in LTE
- **Registration Area:** (NAS) registration area is an area in which the UE may roam without a need to perform location registration, which is a NAS procedure
- When UE registers with the network over the 3GPP access, the AMF allocates a set of tracking areas in TAI List to the UE – called Registration Area.
- Registration area is dynamic
- RA's different for 3GPP and N3GPP. Different TA list for each access.
- Registration area is UE specific

Mobile Initiated Connection Only - (MICO) Mode

- Introduced in 5G
- For IOT Devices that are mostly fixed location
- No Network initiated call/Data connectivity
- No Paging
- If AMF is serving complete PLMN, it may give PLMN as RA for MICO
- MICO Mode: Negotiated by UE and AMF during every registration procedure.

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Features required to support Mobility

- **UE Idle, No connections:**
  - Keep the location of UE as it moves - UE informs Core through Registration requests
  - Contact a UE when required - Paging for UE
  - UE should be able to contact Network to send data while on the move - Service Request

- **UE Connected, Data sessions active:**
  - Keep looking for better Cell to connect
  - Handover to the better Cell when found

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UE in Idle mode (RM-Registered)

- UE select/reselects and camps on a cell (selected PLMN)
- UE listens to any paging from the Network
- UE RRC sends RAN area updates to RAN
- UE NAS sends updates to Core periodically (Periodic Registration) or when moving to a new RA (Mobility Registration)

When UE wants to send data to NW
- UE sends a Service Request

When NW has data to send to UE
- Core pages for UE
- UE sends a Service Request

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Mobility in Connected Mode

RRC Inactive

RRC Connected

UE in RRC-Inactive mode (RM-Registered)

- UE listens to any paging from RAN
- UE RRC sends RAN Notification Area Update to RAN (periodically OR on moving to new RAN Notification area)
  - RAN Notification area is an area where the UE can move without notifying network

When UE wants to send data to NW

- UE sends a RRC Resume request and establishes RRC connection

When NW has data to send to UE

- RAN pages for UE
- UE sends a RRC Resume request and establishes RRC connection

UE in RRC-Connected mode

- Cell level mobility
- Connectivity to change from a NG-RAN node to another
- PDU Sessions are active
- Session and Service Continuity to be ensured
  - SSC Modes introduced in 5G
- Different Levels of mobility in Connected mode
  - Between Radio Units, Distributed Units, gNodeBs, Cores

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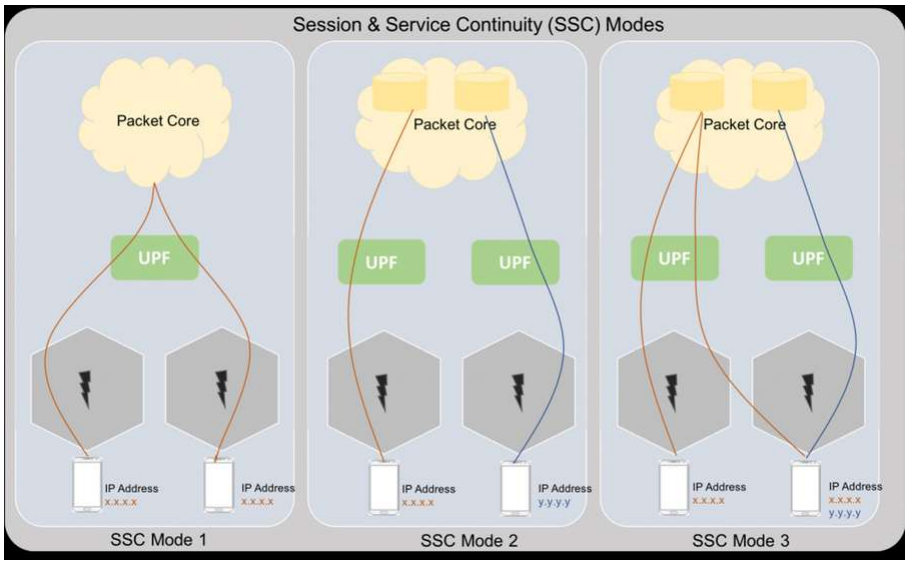
Session Continuity in 5G – SSC mode

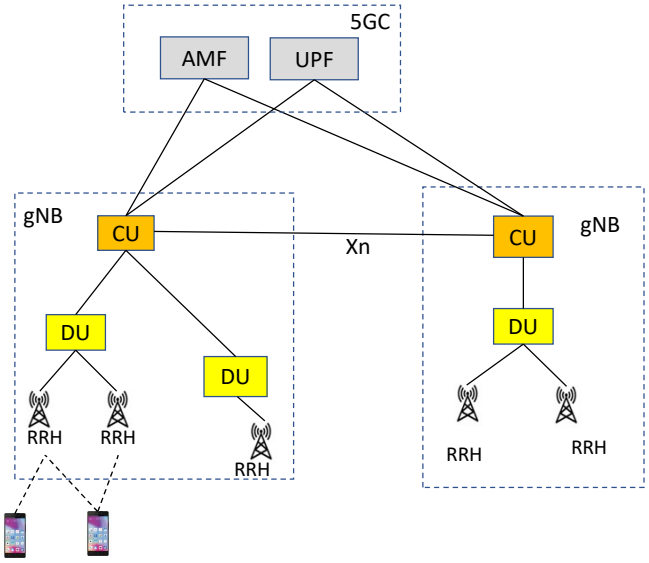
- 4G has Session continuity - anchoring a PDN Connection to a P-GW , wherever the UE moves IP address allocated remains the same
- 5G - Session and Service Continuity – in 5G 3 modes supported.
  - SSC mode 1: same network anchor regardless of device location.
    - An application running on the device will therefore be able to keep using the same IP address on the same interface
  - SSC mode 2: break-before-make (BBM) behavior.
    - When the device leaves the service area of its first network anchor, the network stops using it and starts using a second network anchor closer to the device.
  - SSC mode 3: make-before-break (MBB) behavior.
    - When the device leaves the service area of its first network anchor, parallely attaches to second network
    - no limit set by the 5G for the number of concurrently used network anchors
    - IP address freshly allocated by new Network, parallelly uses multiple IP addresses

SSC Modes in 5G



Source: 4g5gworld.com

Handovers at various levels



1. Intra-gNB-CU Mobility
   a. **Intra-gNB-DU Mobility**
   b. Inter-gNB-DU Mobility
2. Inter-NG-RAN - Xn based
   - With/Without change in UPF
3. Inter-NG-RAN - N2 based

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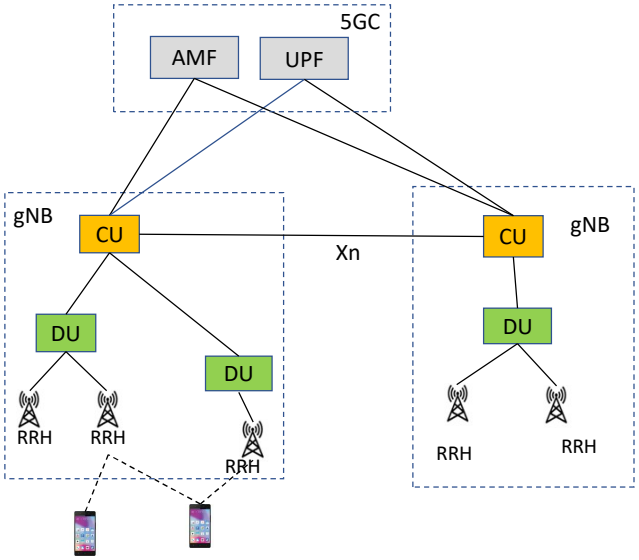

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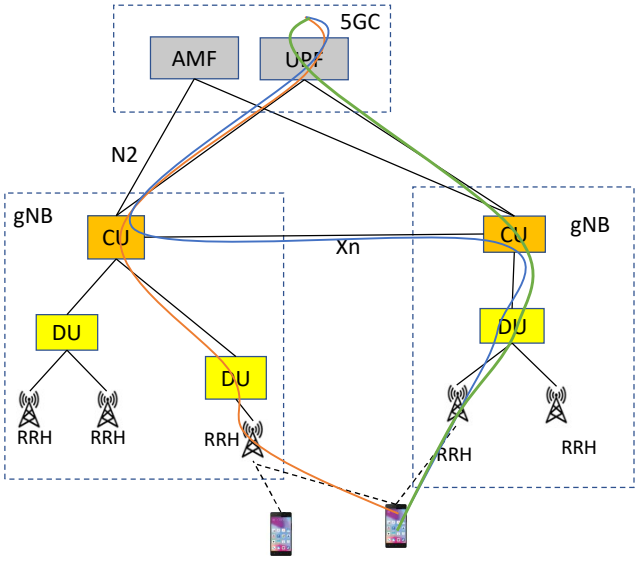

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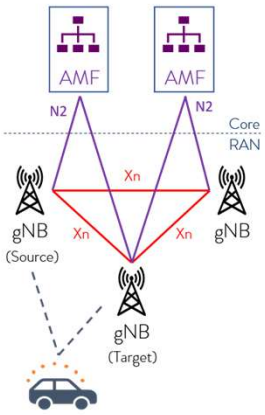
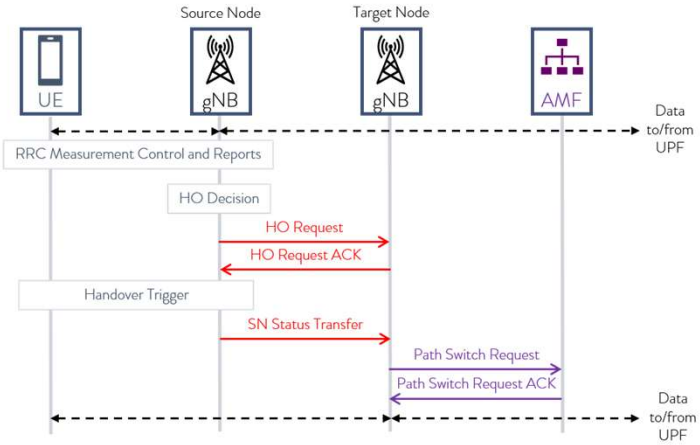
Handover – overall procedure





Summary

UE in CM-Idle mode	Registration Area level	No handover
UE in CM-Connected and RRC Inactive mode	RAN Notification Area level	Context handover
UE in CM-Connected and RRC Connected mode	Cell/gNB level	Session handover

5G Connection & Mobility Management Procedures

UE states

- NAS : MM, SM & CM states
  - Mobility Management state:– UE Core attach state - Registered / De-registered
  - Session Management state: PDU session Active/in-Active. Per PDU session
- Connection Management: Establish & Release Control plane Signaling Connection between UE and AMF
  - Carries NAS signaling messages to AMF
  - Combination of two links
    - RRC signaling between UE and gNB
    - N2-AP signaling between gNB and AMF
  - Maintained by both UE and AMF
- CM-Idle: UE does not have signaling path towards Core Node (AMF) i.e. RRC Idle,
- CM-Connected: UE has signaling connection with AMF i.e. RRC-Connected / RRC-Inactive

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UE states

RM States: UE registration to Core

- RM-Registered
- RM-Deregistered

CM States: AN link

- CM-Idle
- CM-Connected

RRC States: L2/L1 connectivity

- RRC-Idle
- RRC-Connected
- RRC-Connected Inactive

Several Mobility aspects depends on the UE State

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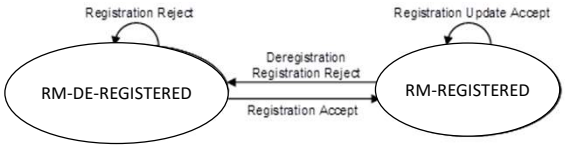
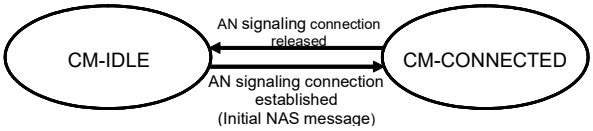
RM/CM states

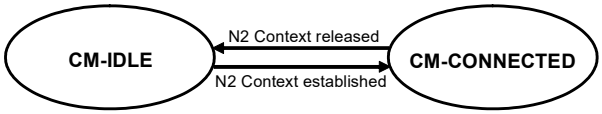
RM State transition in UE and AMF



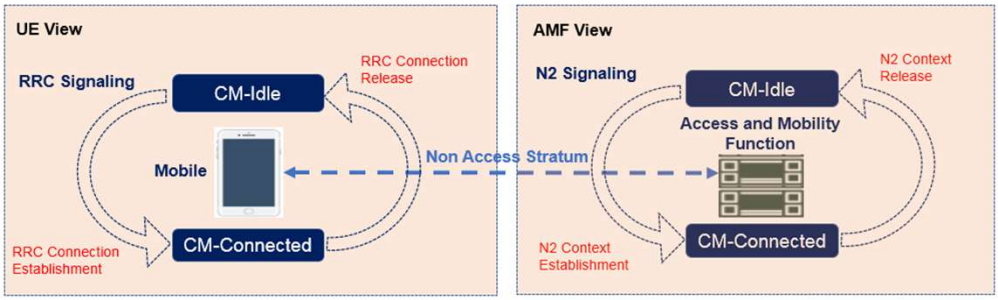
CM State transition in UE



CM State transition in AMF



Connection Management



- UE state moves from CM-IDLE to CM-Connected once RRC connection is established
- AMF UE state moves CM-IDLE to CM-Connected once N2 context is established.
- Typically performed using Registration Request and Service Request.

- UE in CM-CONNECTED, with RRC CONNECTION release, RLF or Link in-activity moves to CM-IDLE
- AMF UE state to CM_IDLE with N2 Context Release: RRC Release, RLF and SCTP link failure

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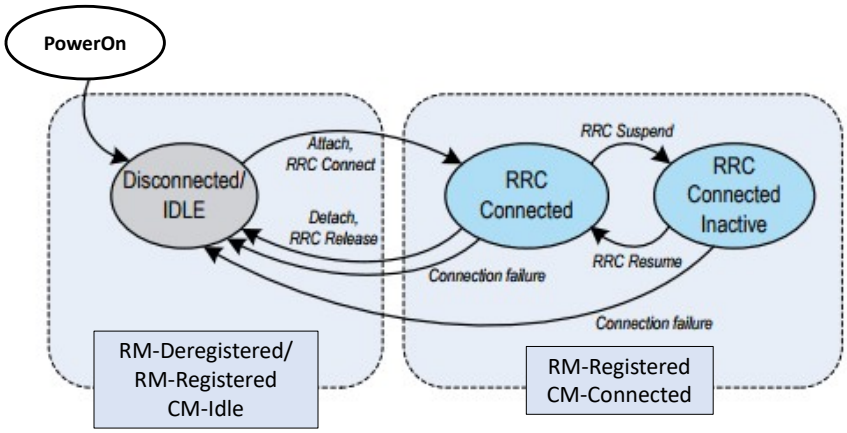
RRC states



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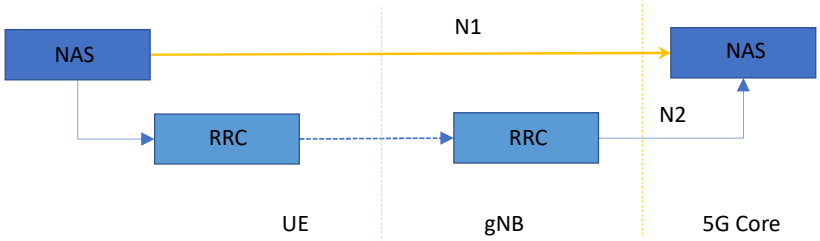
CM state transition

- **UE initiated State Transition – CM-IDLE to CM-CONNECTED**
  - Registration Request and Service Request
  - E.g: UE Initial Registration
  - At Power-on UE selects best gNB based on cell selection procedure
  - To send Registration Request, initiates the RRC connect setup signaling toward gNB
  - N2 Signaling to AMF.
  - Registration Request triggers the transition from CM-Idle to CM-Connected.



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CM state transition

- AMF Initiated : CM-CONNECTED to CM-IDLE
  - AN Release -> N2 context release
  - AMF initiated release of N2 context
  - This should trigger RRC release leading to CM-IDLE on UE too
  - Eg. After Initial UE registration with no PDU sessions to setup, AMF can initiate N2 release
- UE initiated: CM-IDLE to CM-CONNETED
  - For PDU session Establishment, Signaling path needs to be re-established – **Service Request**
  - Periodic / Mobility registration – **Registration Request**
- AMF Initiated: CM-IDLE to CM-CONNECTED
  - Downlink Data / signaling triggers - **Paging**
  - **CN Paging : RRC-IDLE**

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CM state Transition - Conclusion

- For every CM-IDLE to CM-CONNECTED state transitions, UE has to initiate RRC connection setup
- Procedure used at NAS is either Registration Request / Service Request
- NG-AP sets up new N2 context on receipt of these messages and forwards them to AMF using “Initial UE message” that gets AMF to CM-CONNECTED state
- For Network to trigger this transition PAGING is used
- RRC Connection release takes CM-CONNECTED to CM-IDLE

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AN release

- To release logical NG-AP signalling connection, associated N3 User Plane connections, and (R)AN RRC signalling and radio resources.
- Initiated by NG-AP or AMF based on other entity failure detection, RLF, OEM etc
- RAN-initiated causes can be
  - O&M Intervention, Unspecified Failure, RLF, User Inactivity, Inter-System Redirection, request for establishment of QoS Flow for IMS voice, Release due to UE generated signalling connection release, mobility restriction etc.;
- AMF-initiated with cause e.g.
  - reception of Service Request or Registration Request to establish another N1 connection via NG-RAN
  - Unspecified Failure, etc.
- All User plane connections of the associated UE(s) to be disconnected
- N2 UE Context Release Request/Response/Reject Commands are used

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Service Request Procedure

- NAS 5G MM Connection Management procedure
- To change 5GMM mode from CM-IDLE to CM-CONNECTED mode
- UE initiated procedure in CM-IDLE or CM-CONNECTED mode
- Triggers can be
  - For uplink Signaling in CM-IDLE mode
  - When Network has Downlink signaling /Data in CM-IDLE mode
  - Paging : Network/ RAN
  - Notification procedure
  - Fallback indication from RRC
- Initiated separately for each access, i.e 3GPP and non 3GPP
- Followed by N2 context setup and User plane setup procedure (o)

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Service Request Procedure

- User plane connectivity CM-IDLE/CM-CONNECTED
  - PDU session(s) established without User plane
  - To re-establish Data path resources removed by AN
- Can be initiated for both 3GPP and non 3GPP access
  - For access in CM-IDLE, request can come in other CM-CONNECTED access
  - N/w uses Paging if both access are CM-IDLE.
  - No Paging for non 3GPP access
  - N/W uses Notification through CM-CONNECTED access to reach other CM-IDLE access
  - For UE in CM-IDLE over 3GPP access and CM-CONNECTED over non-3GPP access, for DL signaling or user data pending over 3GPP access AMF can trigger either the notification procedure or the paging procedure

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Service Request - Procedure

- Carries Service Type – Signaling/ data / mobile terminates services / emergency services'
- Access type for which SR is sent
- PDU session information: current status, List Of PDU Sessions To Be Activated, List Of Allowed PDU Sessions, security parameters, PDU Session status, 5G-S-TMSI, [NAS message container
- Network can accept / reject based on it's resource status
- The result carries back information on allowed PDU user plane status
- If triggered in response to the paging or NAS Notification, UE sends status List of PDU sessions that can be re-activated
- Optional Authentication procedure from Network if sent without integrity protection

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Notification Procedure

- Network Initiated procedure
  - signaling or user plane activity triggered
  - Sent on other CM-CONNECTED access type
- UE responds with Registration request/Service Request if success
- Service request should carry PDU session status
- On failure UE sends Service Response
- No other parameters sent in Notification Request
- Notification Response can carry PDU session status of other access type that are active

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Paging

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Paging

- When UE is in CM-IDLE state
- Paging is the mechanism in which Network tells UE saying "I have something for you“
- Trigger can be a DL Signaling or a Data in-coming to the UE
- **NAS at AMF will be passed on the trigger from other Appropriate NFs i.e UPF/SMF**
- AMF in-turn forms Paging information and passes through NG-AP to RRC
- gNB RRC in Idle state, Passes on to Lower layer to send Paging information through Paging Physical channels
- UE periodically wakes up and listens to Paging Information transmission channels
- UE RRC Passes on the paging information received to NAS for it to do Service Request/Registration based on the state.

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Types of Paging

- Paging by Core
  - AMF sends paging message to NG-RAN
  - RAN node does paging in the cells that form part of indicated TAI list
- Paging by RAN
  - RAN node does paging in the RNA - may include neighbouring gNBs (Xn based)
  - UE sends request to resume the RRC connection
- Different paging strategies for different types of traffic
  - CM-Idle : AMF applies paging strategy
  - CM-Connected RRC-Inactive: RAN applies paging strategy

- A UE in Idle mode in one access (say 3GPP) can be paged using the other access (say non-3GPP) which is in Connected mode (NAS Notification message)

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Thanks